

Report
Of
BCG GENAI JOB SIMULATION

(A Forage Virtual Experience Program)

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

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Class / Section: 2CSE34

Batch: 2025 – 2029

Academic Year 2026–27

CANDIDATE'S DECLARATION

I, Ritesh Kumar Yadav, a student of B.Tech Computer Science and Engineering (Section 2CSE34, Roll No. 25SCS1003003944) at IILM University, Greater Noida, do hereby declare that this report, titled “BCG GenAI Job Simulation – Financial Data Analysis and AI-Powered Chatbot Prototype,” has been prepared by me on the basis of a self-paced virtual job simulation completed on the Forage platform, modelled on the Generative AI and financial-analytics style of work undertaken by Boston Consulting Group (BCG).

I further declare that this report documents an educational virtual experience programme and does not represent employment, an internship, or any client engagement with Boston Consulting Group. All work described herein, including the financial data extraction, analysis, and chatbot prototype, was completed independently by me using publicly available information and standard open-source tools.

All references used in the preparation of this report have been duly acknowledged, and I have not copied material from any other source without appropriate citation. I affirm that this report is my own work and has not been submitted, in part or in full, to any other institute for the award of any other degree or diploma. I shall be solely responsible for any copyright violation in this regard, if found.

Signature

Student Name : Ritesh Kumar Yadav

Roll No. : 25SCS1003003944

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of the BCG GenAI Job Simulation on the Forage platform. This virtual experience programme gave me an opportunity to gain practical exposure to financial data analysis and simple AI-style application development, modelled on the type of Generative AI and data-driven consulting work undertaken by Boston Consulting Group.

I am thankful to Forage for structuring this job simulation with clear, task-based instructions that allowed me to work through realistic financial-analysis and prototype-development problems in a self-directed manner. The step-by-step task briefs helped me understand a practical approach to extracting and interpreting financial information and translating analytical findings into a working software prototype.

During the simulation, I worked on two connected tasks: the extraction and analysis of financial data from the 10-K filings of Microsoft, Apple, and Tesla, and the development of a rule-based financial chatbot built upon that analysis. Working on these tasks helped me gain practical exposure to data extraction, financial-ratio analysis, Python programming, and simple rule-based application logic.

I would also like to express my sincere gratitude to my teachers and academic mentors at IILM University, Greater Noida, for their continuous support and encouragement, and for providing an academic environment that allowed me to pursue this virtual experience programme alongside my regular coursework. Their guidance helped me maintain a systematic and disciplined approach towards completing this project work.

Finally, I express my heartfelt gratitude to my parents and family for their constant encouragement, support, and blessings, which helped me stay focused and complete this project work successfully.

Signature

Student Name : Ritesh Kumar Yadav

Roll No. : 25SCS1003003944

CERTIFICATE / COMPLETION EVIDENCE

This section is reserved for the inclusion of official completion evidence relating to the BCG GenAI Job Simulation.

As of the date of preparation of this report, a formal completion certificate for this simulation had not yet been issued to the candidate by the Forage platform. In accordance with the requirement that this report should not include invented or assumed certificate details, no certificate image, certificate ID, or issue date has been reproduced here.

Once the official Forage completion certificate for the BCG GenAI Job Simulation is issued and made available, it is intended to be inserted in this section as supporting evidence of the successful completion of both tasks described in this report.



Inspiring and
empowering
future professionals

Ritesh Yadav GenAI Job Simulation

Certificate of Completion
September 17th, 2026

Over the period of September 2026, Ritesh Yadav has completed practical tasks in:

Data extraction and initial analysis
Developing an AI-powered financial chatbot

A handwritten signature in black ink, appearing to read "Tom Brunskill".

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6aaad75bc92b2fc145bad41b | User Verification Code 69463917bf37f9017ae498df | Issued by Forage

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5. PROJECT DESCRIPTION

This report documents the work completed as part of the BCG GenAI Job Simulation, a self-paced virtual experience programme hosted on the Forage platform and modelled on the Generative AI and financial-analytics style of consulting work associated with Boston Consulting Group (BCG). The simulation consisted of two connected tasks: the extraction and analysis of multi-year financial data from public company filings, and the development of a simple, rule-based financial chatbot prototype built upon that analysis.

The work progressed from structured financial data collection and analysis using Python and pandas to the design and testing of a beginner-friendly chatbot capable of answering a defined set of financial questions. Both tasks were completed independently by the candidate, using publicly available SEC filings and standard open-source Python tools, without any real-time data feeds, natural language processing libraries, or external APIs.

This report is organized into subsections covering the introduction, the nature of the BCG GenAI Job Simulation programme, the problem statement, project objectives, scope, technologies used, system architecture, methodology, learning outcomes, and supporting documentation.

5.1.Introduction

Financial analysis is a core activity within management consulting, particularly in engagements that involve evaluating a company's performance, comparing it against peers, or assessing the impact of strategic decisions. With the growing role of Generative AI in business and consulting workflows, understanding how structured financial data can be organized, analyzed, and made accessible through simple software tools has become an increasingly relevant skill for students of computer science and related disciplines.

The BCG GenAI Job Simulation provided an opportunity to engage with this type of work in a structured, task-based format. Rather than focusing purely on theoretical concepts, the simulation required the completion of two practical tasks that, together, illustrated a typical progression in data-driven consulting work: first analyzing raw financial data to generate insight, and then translating that insight into a simple, usable application.

The first task involved extracting Total Revenue, Net Income, Total Assets, Total Liabilities, and Cash Flow from Operating Activities for Microsoft, Apple, and Tesla from their respective Form 10-K filings for fiscal years 2023 to 2025, and analyzing this data using Python and pandas within a Jupyter Notebook. The second task involved building a simple, rule-based Python chatbot capable of answering a defined set of financial questions using the dataset compiled in the first task.

Together, these two tasks provided practical exposure to financial data extraction, data analysis using Python, and the design of simple rule-based logic for an interactive application. The following subsections describe the programme, the problem addressed, the objectives pursued, and the methodology followed in greater detail.

5.2.About BCG GenAI Job Simulation / Forage

Forage is a virtual job simulation platform that partners with organizations across various industries to offer students realistic, self-paced exercises modelled on the type of work performed within those organizations. These simulations are designed to give students practical exposure to industry-relevant tasks without requiring a formal recruitment process, an offer of employment, or direct supervision by the partner organization's staff.

The BCG GenAI Job Simulation is one such programme, offered in partnership with Boston Consulting Group. It is structured around Generative AI and data-driven consulting themes, and is intended to introduce participants to the kind of financial-analysis and AI-application-oriented work that BCG's consulting teams may engage in. The simulation is completed independently by the participant, working through instructions and sample scenarios provided on the Forage platform.

It is important to clarify that completion of this simulation does not constitute an internship, employment, or any client engagement with Boston Consulting Group. No offer letter, stipend, assigned mentor, or fixed commencement and completion dates were issued as part of this programme, and none are claimed in this report. The tasks were undertaken at the candidate's own pace, using publicly available data and standard, freely available software tools.

The programme is nonetheless valuable from an academic perspective, as it exposes participants to a structured, real-world-inspired problem: how financial information disclosed in regulatory filings can be organized and analyzed, and how such analysis might inform the design of simple AI-style tools intended to make financial information more accessible.

5.3.Problem Statement

Publicly listed companies are required to disclose extensive financial information through periodic filings such as the Form 10-K, submitted annually to the U.S. Securities and Exchange Commission (SEC). While this information is comprehensive, it is not readily structured for quick comparison or interpretation. Analysts and consulting teams must manually locate, extract, and organize relevant figures before any meaningful analysis or comparison can take place.

A further complication arises when comparing multiple companies, since fiscal year-end dates are not standardized across all organizations. Microsoft's fiscal year ends in June, Apple's ends in late September, and Tesla follows the calendar year ending in December. Any analysis comparing companies over the “same” fiscal year must therefore account for the fact that the underlying twelve-month periods do not align exactly.

A second, related problem concerns accessibility: even after financial data has been extracted and analyzed, a non-technical user may find it difficult to query that data directly from a spreadsheet or notebook. This suggested a natural extension to the analytical work — exploring whether a simple, rule-based chatbot could allow a user to ask common

financial questions in plain language and receive an answer drawn directly from the analyzed dataset, without needing to open a spreadsheet or notebook themselves.

The overall problem addressed in this project was therefore twofold: first, to extract and analyze multi-year financial data from official filings in a way that accounts for differences such as fiscal year-end dates; and second, to explore, at a prototype level, how such analyzed data could be made more accessible through a simple rule-based conversational interface.

5.4. Project Objectives

The objectives of the BCG GenAI Job Simulation project work were defined around the two tasks undertaken. The major objectives were as follows:

1. To understand how financial information is disclosed in a company's Form 10-K filing and how such information can be located using the SEC EDGAR database.
2. To manually extract Total Revenue, Net Income, Total Assets, Total Liabilities, and Cash Flow from Operating Activities for Microsoft, Apple, and Tesla for fiscal years 2023 to 2025.
3. To compile the extracted financial data into a structured Excel spreadsheet and export it to CSV format for further analysis.
4. To use Python and the pandas library, within a Jupyter Notebook, to calculate year-over-year percentage changes for each financial metric.
5. To compute additional financial ratios, such as net profit margin and debt-to-assets ratio, and to calculate multi-year compound annual growth rates (CAGR).
6. To identify and document meaningful trends and differences in financial performance across the three companies.
7. To account for differences in fiscal year-end dates between Microsoft, Apple, and Tesla when interpreting and comparing the analyzed data.
8. To design a small set of predefined financial questions that a user might reasonably wish to ask about the analyzed dataset.
9. To develop a simple, rule-based Python chatbot using if-elif-else logic that maps each predefined question to an accurate, data-backed response.
10. To implement basic company and fiscal-year detection within the chatbot, along with a sensible default and a clear fallback response for unsupported queries.
11. To test the chatbot with a range of sample and edge-case queries and document the results.
12. To prepare clear, well-organized documentation describing the methodology, findings, and limitations of the completed work.

5.5. Scope of the Work

The scope of this project is limited to two connected tasks undertaken as part of the BCG GenAI Job Simulation. The first task, financial data analysis, and the second task, chatbot prototype development, together define the boundaries of the work described in this report.

Scope of the Financial Data Analysis

The financial data analysis component covers three publicly listed companies — Microsoft, Apple, and Tesla — and five financial metrics: Total Revenue, Net Income, Total Assets, Total Liabilities, and Cash Flow from Operating Activities, for fiscal years 2023 through 2025. The analysis includes year-over-year percentage changes, selected financial ratios, and a written summary of observed trends. The scope does not extend to other companies, additional financial metrics such as earnings per share or segment-level revenue, or periods outside FY2023–FY2025.

Scope of the Chatbot Prototype

The chatbot prototype is scoped to answer five categories of predefined financial questions — total revenue, net income change, total assets, total liabilities, and cash flow from operating activities — for the same three companies and the same three fiscal years used in the analysis task. The chatbot is a rule-based command-line application using if-elif-else logic; it does not include natural language processing, machine learning, external APIs, or live data connections. Any query outside the supported scope is handled through a clearly defined fallback response rather than an attempt at open-ended interpretation.

Overall, the scope of this project is intentionally limited to demonstrate the fundamentals of financial data extraction, structured analysis, and simple rule-based application design, rather than to deliver a production-ready financial analytics or conversational AI system.

5.6. Technologies and Tools Used

The completion of the BCG GenAI Job Simulation tasks involved a small, focused set of programming languages, libraries, and data sources, selected according to the requirements of each task.

5.6.1 Data Sourcing

SEC EDGAR, the U.S. Securities and Exchange Commission's public filing database, was used as the sole source of financial data. The Form 10-K filings of Microsoft, Apple, and Tesla for fiscal years 2023 to 2025 were reviewed to manually extract the required financial figures.

5.6.2 Data Handling and Analysis

- Excel and CSV — used to compile and store the extracted financial data in a structured, tabular format before analysis.
- Python — the primary programming language used for both the data analysis and chatbot development tasks.
- Pandas — used to load the CSV dataset, calculate year-over-year growth rates, compute financial ratios, and perform grouped aggregations by company and year.
- Jupyter Notebook — used as the interactive environment for writing and documenting the analysis, combining code, charts, and narrative markdown

explanations.

- Matplotlib — used to generate comparison charts for revenue, net income, and cash flow from operations across the three companies.

5.6.3 Chatbot Development

- Python dictionaries — used to store the analyzed financial dataset in a structured, lookup-friendly format within the chatbot script.
- if-elif-else logic — used to implement rule-based query matching, mapping recognized keywords to the appropriate data lookup and response.
- Python's input() function — used to provide an interactive command-line interface through which a user can type questions to the chatbot.
- Automated testing (a standalone Python test script) — used to run a batch of sample and edge-case queries and record the chatbot's responses.

No machine learning libraries, natural language processing frameworks, or external web APIs were used in either task. This reflects the intentionally simple, rule-based scope of the chatbot prototype, as defined in the original task instructions.

5.7. System Architecture

Although the two tasks serve different purposes, both follow a structured flow in which raw input is processed in defined stages to produce a meaningful output. The architecture of each task is described separately below.

5.7.1 Financial Data Analysis Architecture

The data analysis task follows a linear pipeline, beginning with the official regulatory filing and ending with documented insight:

SEC 10-K Filing → Manual Data Extraction → Excel / CSV Compilation → Pandas Analysis (pandas + Jupyter) → Insights & Documentation

At each stage, the data becomes progressively more structured: unstructured filing text is converted into discrete numeric figures, those figures are compiled into a tabular spreadsheet, and the spreadsheet is then loaded into pandas for calculation of growth rates, ratios, and comparative charts. The final stage involves documenting observations and conclusions directly within the Jupyter Notebook using markdown narrative.

5.7.2 Chatbot Architecture

The chatbot prototype processes each user query through a sequence of simple, rule-based stages:

User Query → Query (Keyword) Detection → Company / Year Identification → Financial Dataset Lookup → Rule-Based Processing (if-elif-else) → Response

When a user types a question, the chatbot first checks the text for recognized financial keywords, such as “revenue” or “total liabilities.” It then checks for a mentioned company name and fiscal year, applying a default year (FY2025) if none is specified, and prompting for clarification if no company is mentioned. Once a query type, company, and year have been identified, the corresponding figures are retrieved from the financial dataset stored within the script, and a formatted response is returned to the user. If no keyword matches the query, a clearly defined fallback branch returns a message explaining what the chatbot can help with, rather than failing silently or producing an incorrect answer.

This architecture keeps the chatbot's internal logic transparent and easy to trace, which was appropriate given the beginner-friendly, rule-based scope defined for this task.

5.8.Methodology

The project was completed through a structured, stage-by-stage methodology that progressed from requirement understanding to final documentation. The methodology followed for both tasks is described below.

5.8.1 Requirement Understanding

The first stage involved reviewing the task instructions provided through the Forage platform to understand the required financial metrics, companies, fiscal years, and the expected structure of the chatbot prototype.

5.8.2 Financial Data Collection

The Form 10-K filings of Microsoft, Apple, and Tesla were located on the SEC EDGAR database. The required figures — Total Revenue, Net Income, Total Assets, Total Liabilities, and Cash Flow from Operating Activities — were manually identified and recorded for fiscal years 2023 to 2025.

5.8.3 Data Cleaning and Structuring

The collected figures were organized into a structured Excel spreadsheet, with each row representing a company-year combination and each column representing a financial metric. The spreadsheet was then exported to CSV format for use in Python.

5.8.4 Python-Based Analysis

The CSV dataset was loaded into a pandas DataFrame within a Jupyter Notebook. Year-over-year percentage changes were calculated for each metric using grouped operations, and additional derived metrics, including net profit margin, debt-to-assets ratio, and compound annual growth rate, were computed.

5.8.5 Insight Generation

The calculated figures and generated charts were reviewed to identify meaningful trends,

such as differences in growth rates or profitability across the three companies. These observations were documented directly within the notebook using markdown cells.

5.8.6 Chatbot Development

Building on the analyzed dataset, five predefined query categories were defined, and a Python script was developed using if-elif-else logic to map recognized keywords to the corresponding data lookup and response. Company and fiscal-year detection, a default year, and a fallback response were incorporated into the logic.

5.8.7 Testing and Debugging

A separate automated test script was developed to run a batch of sample queries, including both supported and unsupported questions, through the chatbot function. The output was reviewed to confirm that responses were accurate and that the fallback message was returned correctly for unrecognized queries. The results of this testing were saved for reference.

5.8.8 Documentation

Finally, the analytical findings, chatbot functionality, supported queries, and known limitations were documented in a structured, readable format, culminating in this report.

5.9. Learning Outcomes

Completion of the BCG GenAI Job Simulation contributed to the development of the following practical skills and areas of understanding:

- Practical experience in locating and extracting financial figures from real regulatory filings (Form 10-K) via SEC EDGAR.
- Applied proficiency in Python and the pandas library for structured data analysis.
- Understanding of year-over-year growth-rate calculation and basic financial-ratio analysis, including profit margin and CAGR.
- Experience in handling and transforming tabular data using Excel and CSV formats.
- Awareness of the importance of accurate data extraction and data provenance when working with financial information.
- Understanding of how differing fiscal year-end dates can affect cross-company financial comparisons.
- Practical experience designing and implementing rule-based logic using if-elif-else structures in Python.
- Understanding of how a defined dataset can be translated into a simple, queryable chatbot interface.
- Experience in testing and debugging a small software prototype using a dedicated test script.
- Improved technical documentation and communication skills through the preparation of structured project documentation.
- A clearer understanding of the distinction between rule-based systems, such as the

chatbot developed in this project, and true Generative AI systems that rely on natural language understanding and machine learning.

5.10.Documentation / Completion Evidence

The BCG GenAI Job Simulation was completed independently through the Forage platform as a self-paced virtual experience programme. The work completed comprises two tasks: the financial data analysis of Microsoft, Apple, and Tesla, and the development of a rule-based financial chatbot prototype built upon that analysis.

The following project files and records serve as supporting evidence of the completed work:

- Financial data spreadsheet (Excel / CSV) containing the extracted 10-K figures for all three companies across FY2023–FY2025.
- Jupyter Notebook containing the pandas-based analysis, generated charts, and documented observations and conclusions.
- financial_chatbot.py — the complete rule-based chatbot script.
- test_chatbot.py and test_results.txt — the automated test script and its recorded output.
- README.md — documentation describing the chatbot's functionality, supported queries, and limitations.

As noted in the Certificate / Completion Evidence section of this report, a formal Forage completion certificate for this simulation had not yet been issued as of the preparation of this report. No certificate details have therefore been reproduced in this document; the relevant section has instead been reserved for the certificate's inclusion once it is officially issued.

6. REFERENCES

1. U.S. Securities and Exchange Commission, EDGAR Full-Text Search and Company Filings Database — Form 10-K filings of Microsoft Corporation (CIK 0000789019), Apple Inc. (CIK 0000320193), and Tesla, Inc. (CIK 0001318605), sec.gov/edgar.
2. Forage, BCG GenAI Job Simulation — Virtual Experience Programme, task instructions and reference materials, theforage.com.
3. Python Software Foundation, Python Documentation — reference material for Python programming used in the analysis and chatbot development tasks, docs.python.org.
4. Pandas Development Team, pandas Documentation — reference material for data handling, aggregation, and growth-rate calculations, pandas.pydata.org.
5. Project Jupyter, Jupyter Notebook Documentation — reference material for the interactive notebook environment used for analysis and documentation, jupyter.org.
6. Matplotlib Development Team, Matplotlib Documentation — reference material for the comparison charts generated during the analysis task, matplotlib.org.
7. Microsoft Excel and CSV file-format documentation — reference material used for compiling and exporting the extracted financial dataset.
8. Relevant Forage-provided task instructions, requirement briefs, and reference materials used during the completion of the BCG GenAI Job Simulation tasks.